

VJH 2k26 PROBLEM STATEMENTS

Category: Agriculture

Problem Statement 1: Aquaculture Water-Quality and Fish-Health Monitoring System

Description: Develop a smart aquaculture-monitoring system that tracks key water-quality parameters such as temperature, pH, dissolved oxygen, turbidity, and ammonia levels to safeguard fish health. The solution may provide real-time monitoring, anomaly detection, predictive risk alerts, and actionable recommendations to help farmers prevent fish stress, disease, and mortality. This challenge invites innovative applications of IoT, sensor networks, machine learning, and data visualisation to solve real-world aquaculture challenges.

Requirements: Develop a working prototype using real or simulated water-quality sensor data, integrated through an IoT pipeline. The solution should include anomaly detection or predictive analytics to flag unsafe conditions, and present information through a web or mobile dashboard.

Expected Outcomes:

- A working prototype for monitoring key water-quality parameters in real time or via simulation.
- Early-warning alerts and recommendations when water conditions become unsuitable for fish health.
- Visualisation of water-quality trends, current status, and potential risks.
- A user-friendly dashboard demonstrating the potential of smart monitoring in aquaculture.

Problem Statement 2: AR/VR-Based Smart Agriculture Training and Field Assistance System

Description: Develop an AR/VR-based interactive system to support learning, training, and field assistance in agriculture. The solution may provide immersive simulations, interactive 3D models, AR-based identification of crops, pests, or diseases, equipment demonstrations, safety training, step-by-step task guidance, or visualisation of farm information. Students are encouraged to explore innovative applications of AR, VR, 3D modelling, computer vision, spatial computing, and interactive technologies to solve real-world agricultural challenges.

Requirements: Develop an AR/VR or mixed-reality prototype using suitable tools such as Unity, Unreal Engine, ARCore, ARKit, or WebAR. The solution may use agricultural images, videos, or publicly available datasets for computer-vision-based features. It should provide an interactive and user-friendly experience through mobile devices, web platforms, or available AR/VR headsets. Internet connectivity and specialised hardware should not be mandatory; a working prototype or simulation is acceptable.

Expected Outcomes:

- An interactive AR/VR prototype addressing an agricultural use case.
- 3D visualisation, simulation or immersive training experience.
- AR-based crop, pest/disease or equipment assistance, where applicable.
- Interactive step-by-step guidance for agricultural activities.
- A user-friendly application demonstrating the potential of AR/VR in agriculture.

Problem Statement 3: Acoustic AI and Time-Series Anomaly Detection for Grain-Storage Protection

Description: Develop a non-invasive grain-storage protection system that uses acoustic signals and storage-related information to detect hidden insect activity, moisture buildup, fungal risk, and unsafe storage conditions.

Requirements: Use relevant acoustic and grain-storage datasets to identify insect activity, storage abnormalities, and possible quality risks. Apply appropriate audio-analysis and anomaly-detection techniques and present the identified risks, alerts, and recommended actions through a monitoring dashboard.

Expected Outcomes:

- Detection of hidden insect activity using acoustic signals.
- Classification of storage conditions according to risk level.
- Early warning of moisture, fungal, and spoilage risks.
- Generation of a predictive storage-risk score.
- A dashboard displaying storage conditions and recommended actions.

Problem Statement 4: AI-Based Crop and Agricultural Resource Optimization System

Description: Develop an AI-based system that recommends suitable crops and optimizes the allocation of limited agricultural resources. AI should be used to predict crop suitability and expected yield based on soil, weather, and historical production data. The system should determine how available land, water, fertilizer, and budget can be efficiently allocated among the selected crops. The solution should aim to improve expected yield and profitability while reducing water, fertilizer, and resource wastage.

Requirements: Use relevant agricultural datasets to predict suitable crops and their expected performance. Identify important resources and practical constraints involved in crop planning, formulate the crop and resource-allocation task as an optimization problem, and implement an appropriate lightweight AI and classical optimization approach that can run on standard computing systems. Present the optimized recommendations through a simple farmer-friendly dashboard.

Expected Outcomes:

- Recommendation of suitable crops based on soil and weather conditions.

- Prediction of expected yield for shortlisted crops.
- Estimation of water, fertilizer, and cultivation costs.
- Optimal allocation of land, water, fertilizer, and budget.
- Estimation of expected production and profitability.
- Comparison of different resource-allocation strategies.
- A dashboard displaying recommended crops, resource allocation, expected yield, estimated profit, and resource savings.

Problem Statement 5: AI-Based Livestock and Farm Animal Health Monitoring

Description: Develop an intelligent AI system that analyses animal sounds, images/videos, behaviour, and farm conditions to identify potential diseases, stress, abnormal behaviour, and health risks in cattle, buffalo, sheep, goats, poultry, and other animals/birds involved in agriculture and animal production.

Requirements: Use suitable animal audio, image/video, and farm-condition datasets, and apply AI, computer vision, and audio-analysis techniques to identify disease, stress, abnormal behaviour, and unsuitable farm conditions. The system should generate explainable health-risk alerts for farmers and veterinarians and provide a simple monitoring dashboard displaying animal health status and detected risks.

Expected Outcomes:

- Classification of healthy and unhealthy conditions.
- Detection of abnormal sounds and behaviour.
- Monitoring of animal health and farm conditions.
- Early warning and explainable health-risk alerts.
- Real-time animal-health monitoring dashboard.

Category: Climate & Ocean Intelligence

Problem Statement 1: Conversational AI for Weather Forecasting, Alerts, and Climate Information

Description: Weather information is distributed across multiple portals, bulletins, satellite products, and forecasting systems. This makes it difficult for common users, researchers, disaster managers, and government agencies to quickly access relevant and actionable information. Develop an intelligent conversational platform that provides accurate, contextual, and multilingual weather information through natural-language interactions. The platform should support weather forecasts, warnings, climate analysis, and decision-making for different users and application areas.

Requirements: The proposed solution should provide real-time weather information and forecasts through a conversational interface, understand natural-language queries, and generate relevant, location-specific forecasts and advisories. It should communicate extreme-weather alerts and early warnings, support multiple Indian languages, and provide voice-based interaction for improved rural accessibility. The solution should also analyse historical weather information and climate trends, present the results through an accessible and user-friendly interface, and demonstrate accuracy, relevance, responsiveness, scalability, and innovation.

Expected Outcomes:

- A functional conversational AI platform for accessing weather and climate information.
- Faster dissemination of weather forecasts, alerts, and advisories.
- Improved public accessibility to reliable weather information.
- Better preparedness and response during extreme-weather events.
- Location-specific and multilingual weather assistance.
- Decision support for agriculture, aviation, marine operations, disaster management, research, and urban planning.

Problem Statement 2: National Weather Big Data Analytics Platform

Description: Weather-related information is generated continuously through multiple online platforms, public information systems, and community reports. The large volume and varying reliability of this information make it difficult to identify genuine reports and obtain timely insights about weather events. Develop a scalable National Weather Big Data Analytics Platform that collects, processes, verifies, categorizes, and visualizes real-time weather-related information. The platform should help authorized users monitor weather events, identify unreliable or duplicate reports, and analyse location- and time-based patterns through an interactive interface.

Requirements: The proposed solution should collect and process real-time weather information from multiple sources and organize it with relevant contextual details. It should support large-scale data ingestion, processing, storage, and visualization using AI and Machine Learning techniques to identify fake, misleading, unreliable, or duplicate reports. The system should verify the collected information, track its verification status, and automatically categorize different types of weather events. It should also provide date-wise, event-wise, and location-wise analysis and present real-time information and analytical insights through a user-friendly web-based dashboard with suitable monitoring and administrative functions.

Expected Outcomes:

- A scalable platform for collecting and analysing real-time weather information.
- Organized storage and management of weather-related information.
- AI-assisted identification of unreliable and duplicate reports.
- Automatic classification of weather events.
- Verification-status tracking for collected information.
- An interactive dashboard supporting filtering, monitoring, visualization, and real-time analytics.

Problem Statement 3: AI/ML-Based System for Tropical Cyclone Pattern Analysis

Description: Develop an Artificial Intelligence (AI) and Machine Learning (ML)-based system for the identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data. The system should analyse satellite observations from different sources to detect cyclone characteristics such as formation, structure, movement, and intensity. By combining historical and real-time satellite data with suitable AI/ML techniques, the solution should identify recurring cyclone patterns and provide predictions about their future behaviour, supporting early warning and disaster preparedness.

Requirements: The proposed solution should use multi-source satellite data and suitable Artificial Intelligence and Machine Learning techniques to analyse tropical cyclones. It should identify and classify different tropical cyclone patterns and predict their behaviour based on the available satellite observations.

Expected Outcomes:

- An AI/ML-based system for analysing tropical cyclones.
- Identification of different tropical cyclone patterns.
- Classification of tropical cyclone patterns.
- Prediction of tropical cyclone patterns using multi-source satellite data.

Problem Statement 4: AI-Based Reconstruction of Subsurface Ocean Temperature

Description: Subsurface ocean temperature is important for understanding ocean circulation, heat content, climate variability, air-sea interactions, and marine ecosystems, with applications in marine heatwave monitoring and fisheries. However, observations from floats, buoys, gliders, and ships have limited spatial and temporal coverage. Develop an AI and Deep Learning-based system that uses continuous surface satellite observations to learn subsurface-ocean patterns and reconstruct three-dimensional, depth-wise temperature profiles over the North Indian Ocean.

Requirements: The proposed solution should collect, preprocess, and harmonize suitable multi-source satellite and ocean observations and use relevant surface-ocean variables to estimate subsurface temperatures at multiple standard depth levels. It should apply appropriate AI or Deep Learning techniques to learn relationships between surface conditions and subsurface temperature profiles, generate continuous spatial and temporal temperature estimates, and evaluate the reconstructed results using independent observations and suitable performance measures.

Expected Outcomes:

- A functional pipeline for processing multi-source satellite and ocean observations.
- An AI-based model for estimating depth-wise subsurface-ocean temperature.
- Continuous three-dimensional temperature reconstruction across multiple depth levels.
- Evaluation of reconstructed temperatures using independent observations.
- Visual presentation of reconstructed subsurface temperature profiles.

Problem Statement 5: AI/ML-Based Detection and Mapping of Floating Marine Plastic Debris

Description: Marine plastic pollution is a growing environmental challenge, as plastics released into the ocean disperse widely through winds, currents, and waves. The MSC ELSA 3 incident is a real-world example: the Liberian-flagged vessel, travelling from Vizhinjam to Kochi Port, developed a severe starboard list on 24 May 2025 and sank on 25 May 2025, about 13 nautical miles off Kerala. Lost containers included plastic resin pellets (nurdles), later reported along Kerala and southern Tamil Nadu coasts, including Kanyakumari and Dhanushkodi/Rameswaram. INCOIS activated SARAT and the Oil/Nurdles Spill Trajectory System to estimate drift and dispersion using oceanographic and meteorological information, highlighting the need for rapid debris identification, localisation, monitoring, prioritisation, and targeted removal.

Requirement: Develop an AI/ML-based system using open-access, high-spatial- and spectral-resolution satellite observations with oceanographic and meteorological model products to detect, classify, and map floating marine plastic debris. The system should identify potential debris, estimate its location, spatial extent, and detection confidence. INCOIS model products may provide currents, winds, and drift patterns to improve identification of accumulation zones and prediction of likely movement. The solution should transform satellite and model data into actionable intelligence for field verification and targeted marine debris clean-up.

Expected Outcomes:

- AI/ML-based detection and classification of potential floating marine plastic debris from satellite observations.
- Geo-referenced mapping of detected debris, including hotspots and estimated spatial extent/concentration.
- Confidence score or probability associated with each detection.
- Integration of oceanographic and meteorological data to estimate debris drift, dispersion, and potential future accumulation zones.
- Generation of a priority map identifying areas suitable for field verification and targeted clean-up.
- An interactive GIS-based dashboard displaying satellite imagery, detected debris, hotspots, confidence, and relevant environmental information.
- Demonstration of the proposed solution using the MSC ELSA 3 incident as a real-world case study, wherever suitable data are available.

Category: Healthcare

Problem Statement 1: AI-Based Early Warning System for Patient Health Deterioration

Description: Timely identification of patient deterioration is critical in hospitals. Changes in vital signs may indicate the onset of serious complications, but continuous manual monitoring can be challenging, especially when healthcare professionals manage multiple patients. Develop an AI-powered Early Warning System that analyses patient health parameters such as heart rate, SpO₂, temperature, blood pressure, and other available clinical observations to identify patterns indicating a potential risk of deterioration. The system should generate an interpretable risk score and timely alerts to help healthcare professionals prioritise patients who may need immediate attention.

Requirements: The proposed solution may include real-time or simulated patient monitoring, AI/ML-based risk prediction, patient risk scoring, early-warning alerts, a visual dashboard for healthcare professionals, and explainable factors contributing to the prediction. Suitable technologies may include Machine Learning, Deep Learning, Time-Series Analysis, IoT, Edge Computing, Explainable AI, and a web- or mobile-based dashboard.

Expected Outcomes:

- An AI-based system for identifying patients at risk of health deterioration.
- An interpretable risk score based on available patient-health parameters.
- Timely early-warning alerts for healthcare professionals.
- A visual dashboard for monitoring and prioritising patients.
- Explanations of the factors contributing to risk predictions.
- Support for faster clinical intervention and improved patient safety.

Problem Statement 2: Out-of-Pocket Healthcare Expenditure Prediction and Reduction

Description: High out-of-pocket healthcare expenditure remains a major concern for Indian families. Patients may not know the approximate costs of treatments and diagnostic procedures or be aware of alternative healthcare facilities. Develop an AI-based Healthcare Cost and Care Navigation Platform that provides an estimated treatment cost range, identifies appropriate nearby healthcare facilities, and highlights potentially available public health schemes or insurance coverage.

Requirements: The proposed solution should estimate the probable cost range of treatments or diagnostic procedures, recommend appropriate nearby healthcare facilities, and provide relevant information about potentially applicable public health schemes or insurance coverage. The platform may use Artificial Intelligence, recommendation systems, knowledge graphs, and GIS to generate suitable cost and care-navigation recommendations through an accessible interface.

Expected Outcomes:

- Estimated cost ranges for treatments and diagnostic procedures.
- Recommendations for appropriate nearby healthcare facilities.
- Information about potentially available public health schemes or insurance coverage.
- A healthcare cost and care-navigation platform for patients.
- Better-informed healthcare decisions and a potential reduction in patients' financial burden.

Problem Statement 3: Patient-Held Consent-Driven Health Record

Description: A patient may visit a Primary Health Centre (PHC), private clinic, and district hospital, with each provider starting the consultation without access to earlier medical information. This can result in repeated tests, limited awareness of previously prescribed medicines, and missed information about allergies recorded elsewhere. Although national digital-health infrastructure exists, there are limited solutions that small healthcare providers can

easily adopt. Develop a patient-held health record that enables providers to quickly access relevant medical history at the point of care while giving patients control over access to their information.

Requirements: The proposed solution should provide a patient-held, consent-driven health record that is QR-based, ABHA-linked, or supports both approaches. It should allow patients to grant healthcare providers access to relevant medical history and revoke that access when required. Both consent and revocation must be clearly demonstrated in the prototype and should not be limited to a checkbox in a settings screen. The provider should be able to access the required information within seconds using a simple interface that does not depend on a laptop, a complex login process, or extensive training.

Expected Outcomes:

- A patient-held digital health record accessible through QR code, ABHA linkage, or both.
- Quick access to relevant medical history at the point of care.
- Reduced repetition of medical tests and prescriptions.
- Improved visibility of previously prescribed medicines and recorded allergies.
- Demonstrable patient-controlled consent and access revocation.
- A simple solution suitable for adoption by small healthcare providers.

Problem Statement 4: AI-Based Detection of Medication Errors and Dangerous Drug Interactions

Description: Medication errors and harmful drug interactions can occur when prescriptions are issued without considering a patient's existing medications, allergies, diagnoses, and other clinical information. Develop an intelligent system that analyses this information to identify potentially dangerous drug-drug interactions, drug-disease interactions, duplicate medications, incorrect dosages, and allergy conflicts.

Requirements: The proposed solution should analyse prescriptions, patient medications, allergies, diagnoses, and available clinical information to identify possible medication-safety

risks. It may use Natural Language Processing, knowledge graphs, Large Language Models, Graph Neural Networks, or a hybrid approach combining rule-based methods with Machine Learning. The system should provide an explainable safety warning describing the identified risk instead of displaying only a general “interaction detected” message.

Expected Outcomes:

- Detection of potentially dangerous drug-drug interactions.
- Identification of drug-disease interactions.
- Detection of duplicate medications and incorrect dosages.
- Identification of conflicts involving patient allergies.
- Explainable safety warnings describing the detected risk.
- An intelligent medication-safety support system for healthcare professionals.

Problem Statement 5: AI-Based Hospital Resource Optimization

Description: Hospitals must efficiently manage limited resources while responding to changing patient demand. Develop an AI-based system that predicts hospital resource requirements and recommends the optimal allocation of beds, ICU capacity, operating rooms, staff, and emergency resources under available operational constraints.

Requirements: The proposed solution should analyse available hospital-resource and demand information, forecast future requirements, and recommend suitable resource allocation under defined constraints. It may use time-series forecasting, optimization methods, Reinforcement Learning, Operations Research, and predictive analytics to support effective planning and institutional decision-making.

Expected Outcomes:

- Prediction of hospital-resource demand.
- Optimized allocation of beds and ICU capacity.
- Improved scheduling and utilization of operating rooms.
- Effective allocation of staff and emergency resources.

- Resource-allocation recommendations under operational constraints.
- A decision-support interface for hospital administrators.
- Improved hospital resource utilization and real-world institutional impact.

